

Genotypes that change prescription

Gene	Genotype	Phenotype	Trust
CYP2C9	*1/*3 (rs1057910 A/C; AS 1.0)	Intermediate metabolizer	High (chip)
VKORC1	−1639 T/T (rs9923231)	High sensitivity to coumarins	High (Imputed DR2 0.99 + chip concordant)
NAT2	*5/*5	Slow acetylator	High (chip, 7 SNPs)
CYP3A5	*1/*3	Expression (more debugging)	High (chip)
UGT1A1	*1/*28	Gilbert (glucuronidation something smaller)	High (chip)
CYP2D6	Probable *1/*2	Probable normal; number of undetermined copies	Moderate
CYP2B6	Probable *2/*4	Fast metabolizer	Moderate
HLA-B*58:01	Probable negative (imputed tag)	Tip before allopurinol	Moderate

Avoid or adjust

Situation	Indication (suggestion, not applied)	Gene
NSAID	Avoid piroxicam and tenoxicam. Meloxicam: 50% of the dose. Ibuprofen, celecoxib, flurbiprofen, lornoxicam: lower dose, short course. Avoid chronic NSAID.	CYP2C9 *1/*3
Anticoagulation	Prefer direct anticoagulant (apixaban, rivaroxaban, edoxaban, dabigatran). If warfarin/acenocoumarol: very low initial dose by genetic algorithm and tight INR.	CYP2C9 *3 + VKORC1 T/T (CYP4F2 C/C)
Phenytoin/fosphenytoin	~25% less maintenance, guided by levels.	CYP2C9
Siponimod	Maintenance 1 mg/day.	CYP2C9 *1/*3
Fluvastatin	≤ 40 mg/dia.	CYP2C9
Sulfonyleureas	Start with low dose and monitor blood glucose.	CYP2C9
Isoniazid	NAT2-guided dose and hepatic surveillance.	Slow NAT2
Hydralazine	40–75 mg/day with care.	Slow NAT2
Sulfasalazine, sulfamethoxazole, dapsone, procainamide, amifampridin	Monitoring adverse effects; amifampridin at the lowest dose.	Slow NAT2
Tacrolimus	Initial dose 1.5–2× with levels.	CYP3A5 expressor
Sirolimus	Calibrate the dose with 24–48 h after a dose.	CYP3A5 expressor
Irinotecan, sacituzumab	Monitoring neutropenia. Nilotinib and pazopanib: higher bilirubin (benign).	UGT1A1 *28
Paracetamol	Moderate dose (maximum suggested 2 g/day), due to liver genetics.	UGT1A1 *28
Codeine, tramadol, tamoxifen, eliglustate, pmozide	Standard dose only after confirming the CYP2D6 with copy number.	CYP2D6
Allopurinol	HLA-B*58:01 typing before starting.	HLA
Methadone	Possible lower levels; titer by response and ECG (QT).	Fast CYP2B6
Methotrexate	Support with 5-MTHF (MTHFR heterozygous compound).	MTHFR
Aminoglycosides	Short course and auditory surveillance (non-determinable MT-RNR1).	MT-RNR1
Pioglitazone	Not recommended for genetic reasons: High bladder cancer score, slow NAT2 and high reduced ejection fraction score.	Several

Safe in standard dose (normal genotype)

Drugs	Gene
Clopidogrel, PPI (omeprazole, pantoprazole), citalopram, escitalopram, sertraline	CYP2C19 *1/*1 (normal)
All statins at the desired dose (except fluvastatin ≤ 40 mg)	Probable SLCO1B1 *1/*14 (augmented function; without the risk allele rs4149056), normal ABCG2
5-fluorouracil, capecitabine	Normal DPYD (13 variants; repeat clinical DPD, as EMA recommends)
Azathioprine, mercaptopurine, thioguanine	TPMT *1/*1, NUDT15 *1/*1
Primaquine, tafenoquine, rasburicase, dapsone, methylene blue	Normal G6PD (hemizygous)
Abacavir	HLA-B*57:01 negative
Carbamazepine	HLA-A*31:01 probable negative (confirm if long therapy; B*15:02 non-evaluable)
CYP3A4 substrates (normal genome; interactions with strong inhibitors, such as oral azoles with sirolimus, remain)	CYP3A4 *1/*1

Anesthesia and surgery

Topic	Genotype	Indication (suggestion, not applied)
Malignant hyperthermia	RYR1 (34 SNPs) and CACNA1S (2) without variants	Reduced risk, not absent: standard precautions with volatiles and suxametonium
Succinylcholine, mivacurium	Normal BCHE (without D70G or A539T)	Normal duration
Opioids	OPRM1 A/G, COMT Val/Val	Need a little higher; holder of the answer
Postoperative nausea	CHRM3 (1 allele)	Prophylaxis with probable normal ondansetron (CYP2D6) and dexamethasone
Thrombosis	Venous thrombosis score p86.3; F5 Leiden and prothrombin negative	Mechanical prophylaxis and/or low molecular mass heparin according to surgery; early mobilization

Topic	Genotype	Indication (suggestion, not applied)
Analgesia	CYP2C9 *1/*3	Short-term NSAID in minimal dose

Other genes tested (no action)

Gene	Result
CYP2C19	*1/*1 (without *2, *3, *4, *17): normal metabolizer
CYP4F2	*1/*1 (rs2108622 C/C)
CYP1A2	*1F heterozygous (intermediate inducibility; weak clinical effect)
SLC22A1 (OCT1)	Normal (morphine, tramadol, ondansetron, metformin)
IFNL3 / IFNL4	Genotype of favourable response to interferon
POLG	No variants (valproate without genetic restriction)
F5 Leiden, prothrombin G20210A	Negative (estrogens, tamoxifen, thrombopoietic)
HLA-B*15:02, HLA-A*33:01	Not assessable in the chip

Notes

- **What to confirm with clinical test:** CYP2D6 with copy number and HLA typing of 2 fields (B*58:01, A*31:01, B*15:02).
- **Current medication:** None of them has actionable pharmacogenetic guidelines; sirolimus is calibrated by dosage and isotretinoin asks for hepatic surveillance.
- This card summarizes the Pharmacogenomic and Anesthesia reports and Surgery. It is not a prescription: any decision is made by the doctor.